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SPEAKER

Abstract

Provided is a speaker capable of presuming a cause of failure. A magnet, a vibrator, a voice coil configured to vibrate the vibrator, and a detector insulated from the voice coil and including a closed circuit of a conductor that is fused under a predetermined fusion condition, and the detector is disposed at a position where power is generated due to fluctuation in a magnetic field generated by the voice coil.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese patent application No. 2024-030587, filed on Feb. 29, 2024, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

[0002] The present disclosure relates to a speaker.

[0003] A voice coil in a speaker generates heat or could be burnt when an excessive current is input to the voice coil.

[0004] Patent Literature 1 describes a speaker in which a drive current is made to flow through a voice coil via a fuse.

[0005] Further, Patent Literature 2 describes a speaker in which a breaker or a fuse is connected to a coil.

[0006] Patent Literature 1 Japanese Unexamined Patent Application Publication No. 2004-320264

[0007] Patent Literature 2 Japanese Unexamined Patent Application Publication 2008-259247

[0008] However, there has been a problem in that when a voice coil of a speaker fails by a burnout, it is impossible to identify and presume whether the cause of the burnout is an excessive input or a factor other than the excessive input. Further, in the speakers of Patent Literature 1 and Patent Literature 2, there has been a problem in that when a fuse blows, the speaker itself can no longer function, and thus, the cause of failure of the speaker is difficult to identify.

SUMMARY

[0009] A speaker of one embodiment includes: a magnet; a vibrator; a voice coil configured to vibrate the vibrator; and at least one detector insulated from the voice coil and including a closed circuit of a conductor that is fused under a predetermined fusion condition, in which the at least one detector is disposed at a position where power is generated due to a fluctuation in a magnetic field generated by the voice coil.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other aspects, advantages and features will be more apparent from the following description of certain embodiments taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1 is a cross-sectional view showing a schematic configuration of a speaker according to Embodiment 1;

[0012] FIG. 2 is a perspective view showing the schematic configuration of the speaker according to Embodiment 1; and

[0013] FIG. 3 is a perspective view showing the schematic configuration of the speaker according to Embodiment 1.

DETAILED DESCRIPTION

[0014] For clear explanation, the following descriptions and drawings are omitted and simplified, as appropriate. In the drawings, the same elements are assigned the same reference signs, and overlapping descriptions are omitted, as necessary. Further, for clear explanation, the descriptions will be made using an XYZ three-dimensional orthogonal coordinate system. Relative to a speaker **100**, a front-rear direction (depth direction), a left-right direction (horizontal direction), and an up-down direction (vertical direction) are referred to as a Z-direction, an X-direction, and a Y-direction, respectively. A front direction, a rear direction, a right direction, a left direction, an upward direction, and a downward direction are referred to as a +Z-direction, a -Z-direction, a +X-direction, a -X-direction, a +Y-direction, and a -Y-direction, respectively.

Embodiment 1

[0015] Hereinafter, with reference to the drawings, an embodiment of the present disclosure will be described. FIG. 1 is a cross-sectional view showing a schematic configuration of a speaker according to Embodiment 1. In FIG. 1, the speaker **100** includes a magnet **101**, a vibrator **102**, a voice coil **103**, a detector **104**, a frame **105**, a dumper **106**, an edge **107**, a plate **108**, a yoke **109**, and a cap **110**.

[0016] The magnet **101** is, for example, a permanent magnet.

[0017] The vibrator **102** is physically connected to the voice coil **103**. Due to the vibration of the vibrator **102**, air vibration (compressional wave) is generated. For example, the vibrator **102** has a ring shape. Further, a cross-sectional shape of the vibrator **102** may be in any shape, and may be in a curved shape as shown in FIG. 1. Furthermore, the vibrator **102** is preferably composed of a synthetic resin, paper, cloth, or the like.

[0018] The magnet **101** forms a magnetic field closed circuit through interaction between the magnet **101** and iron materials such as the plate **108** and the yoke **109**. The voice coil **103** vibrates in the front-rear direction (Z-direction) of the speaker **100** upon receipt of an electromagnetic force from a magnetic field formed in a gap around the magnetic field closed circuit, in response to change with time in the current of an electric signal applied to the voice coil **103**. The vibrator **102** also vibrates in the front-rear direction (Z-direction) of the speaker **100** due to the vibration of the voice coil **103**. The speaker **100** can emit sound due to this vibration.

[0019] The detector **104** includes a closed circuit of a conductor that is fused under a predetermined condition. In the following descriptions, the condition under which the detector **104** is fused may be referred to as a fusion condition. The detector **104** is, for example, a conductor formed in a doughnut shape. The detector **104** may be a conductor with a coil wound in a helix or a spiral shape a plurality of turns, as long as it is a closed circuit with a start point and an end point connected. Here, the detector **104** is preferably formed in a round shape, but not limited thereto, and may be in an elliptic or a quadrangular shape. The detector **104** is insulated from the voice coil **103**. The detector **104** is fixed at a position where power is generated due to fluctuation in the magnetic field generated by power applied to the voice coil **103**. The detailed principle of detection by the detector **104** will be described later.

[0020] For example, the detector **104** is disposed such that a center axis of the voice coil **103** is located on an inner side the closed circuit of the detector **104**. More specifically, the detector **104** is preferably disposed such that the center axis of the voice coil **103** is located near a center axis of the detector **104**. Here, the center axis of the detector **104** is a center axis of the speaker **100**, and the center axis of the voice coil **103** is a center axis of a loop formed by a winding of the closed circuit of the detector **104**.

[0021] For example, the detector **104** is disposed such that an angle of a plane in which the voice coil **103** is wound and an angle of a plane formed by the winding of the closed circuit of the detector **104** are coincide with or proximate to each other. In other words, the plane on which the voice coil **103** winds and the plane formed by the closed circuit winding of the detector **104** are approximately parallel. For example, as in FIG. 1, when the plane in which the voice coil **103** is wound is an XY-plane, the plane formed by the closed circuit of the detector **104** is also the XY-plane.

[0022] For example, the detector **104** may be fixed on an exterior surface of the speaker **100**, that is, a side that is visually recognized by a user of the speaker **100**. The detector **104** is fixed on the exterior surface of the speaker **100** so that it can be presumed that a failure was caused by application of an excessive power to the coil, by visual observation without disassembling the speaker. For example, the detector **104** is fixed on a surface of the cap **110** as in FIG. 1. Further, the detector **104** may be disposed on a site other than the exterior surface of the speaker **100**. The detailed arrangement of the detector **104** will be described later.

[0023] The fusion condition of the detector **104** can be set in any manner in accordance with the positional relation between the detector **104** and the voice coil **103**, and a thickness and a resistance

value of a conductor wire of the closed circuit. The detailed fusion condition of the detector **104** will be described later. Further, the speaker **100** may include a plurality of detectors **104** having different fusion conditions.

[0024] The frame **105** is a frame that supports each component of the speaker **100**.

[0025] The dumper **106** physically connects the voice coil **103** and the frame **105** and holds the voice coil **103** moving in the up-down direction (Z-direction) at an accurate position.

[0026] The edge **107** is physically connected to the vibrator **102** and the frame **105** and holds the vibrator **102** at an accurate position.

[0027] Next, the principle of the detector **104** will be described. FIG. 2 is a perspective view showing the schematic configuration of the speaker according to Embodiment 1. In this example, unlike FIG. 1 in which the detector **104** is disposed on the exterior surface of the speaker **100**, as shown in FIG. 2(a), an example in which the detector **104** is disposed in an upper portion (+Z side) of the yoke **109** other than the exterior surface of the speaker **100** will be described.

[0028] When current flows to the voice coil **103**, a magnetic field is generated in a direction perpendicular to the plane in which the voice coil **103** is wound. Specifically, the magnetic field is generated in the up-down direction (Z-direction) as denoted by an arrow in FIG. 2(b). The arrow in FIG. 2(b) also shows the variation of the magnetic field generated by the current flowing in the voice coil **103**, and the arrow penetrates the plane formed by the closed circuit of the detector **104**. In the voice coil **103**, since the current changes corresponding to a sound to be generated, this magnetic field also changes.

[0029] Due to the change in the magnetic field generated by the voice coil **103**, electromagnetic induction occurs in the detector **104**, so that the current flows to the detector **104** as denoted by an arrow in FIG. 2(c). In other words, the current flows through the detector **104** by placing the detector **104** in a position where the variation of the magnetic field generated by the current flowing in the voice coil **103** passes through the plane formed by the closed circuit of the detector **104**.

[0030] The value of the current flowing to the detector **104** fluctuates in proportion to the magnitude of the fluctuation in the generated magnetic field. Therefore, the value of the current flowing to the detector **104** (in other words, a power value to be added) is in proportion to the magnitude of an electric signal input to the voice coil **103**.

[0031] The fusion condition of the detector **104** can be set based on the relation between the electric signal input to the voice coil **103** and the value of the current (or the power value) generated in the detector **104** by the electromagnetic induction. That is, it is only necessary to set the detector **104** so as to be fused at the value of the current (or the power value) generated in the detector **104** corresponding to the power value input to the voice coil **103** to be detected.

[0032] In the following description, the power value input to the voice coil **103** to be detected may be referred to as a detection power value. The detection power value may be, for example, a maximum power value that is allowed to be input to the voice coil **103**. That is, the fusion condition is set to be the value of the power generated in the detector **104** corresponding to the maximum power value that is allowed to be input to the voice coil **103**, so that it can be presumed that when the speaker **100** fails, in a case where the detector **104** is fused, the cause of failure is a burnout due to an excessive input, and in a case where the detector **104** is not fused, the cause of failure is another factor. The cause of failure other than the excessive input is, for example, mechanical failure.

[0033] When the speaker **100** includes a plurality of detectors **104**, the detectors **104** may be set so as to be fused under different fusion conditions. For example, a case is assumed in which the fusion condition of a given detector **104** is set to be a value of the power generated in the detector **104** corresponding to a maximum power value that is allowed to be input to the voice coil **103**. At this time, the fusion condition of a detector **104** different from the given detector **104** is a value of the power smaller than the fusion condition of the given detector **104**. In this manner, the speaker **100**

includes a plurality of detectors **104** for which different fusion conditions are respectively set, so that the maximum power value input to the voice coil **103** can be presumed in more detail.

[0034] The maximum power value input to the voice coil **103** can be presumed in more detail, so that for example, also in a case where the speaker **100** is not failed, whether the detector **104** is fused can be confirmed at the time of a vehicle inspection. If none of the plurality of detectors **104** is fused, it can be presumed that only power values lower than the detection power values corresponding to the plurality of fusion conditions are input to the voice coil **103**. If one or more of the plurality of detectors **104** are fused, it can be presumed that power values corresponding to the fusion conditions of the fused detectors **104** were input to the voice coil **103**. In this manner, the use condition of the speaker on the market can be identified, which can be used as a reference for design values of the allowable power for the speaker.

[0035] Next, the position of the detector **104** will be described. FIG. **3** is a perspective view showing the schematic configuration of the speaker according to Embodiment 1. In FIG. **3**, in the speaker **100**, the position where the detector **104** is disposed is **104-1**, **104-2**, **104-3**, and the like.

[0036] The detector **104** may be disposed at a position denoted by **104-1**, that is, on a surface of the edge **107** and the exterior surface of the speaker **100**. When the detector **104** is disposed at the position **104-1**, since it can be disposed without interfering with other components of the speaker **100**, there is an advantage in that the production of the speaker **100** is less affected.

[0037] The detector **104** may be disposed at a position denoted by **104-2**, that is, on the surface of the cap **110** and the exterior surface of the speaker **100**. The position denoted by **104-2** is the same as the position of the detector **104** shown in FIG. **1**. When the detector **104** is disposed at the position **104-2**, it is disposed on an appearance surface of the speaker **100**, so that it becomes easier to confirm whether the detector **104** is fused.

[0038] The detector **104** may be disposed at a position denoted by **104-3**, that is, a position in a vicinity of the magnet **101** other than the exterior surface of the speaker **100**. The position of the detector **104** denoted by **104-3** is, for example, the upper portion (+Z side) of the yoke **109**, but the configuration is not limited thereto, and the detector **104** may be disposed in an upper portion (+Z side) of another iron material, such as the plate **108**. The position denoted by **104-3** is the same as the position of the detector **104** shown in FIG. **2**. When the detector **104** is disposed at the position **104-3**, since the detector **104** is close to the voice coil **103**, it is possible to increase the value of the power generated in the detector **104** when the same power value is input to the voice coil **103** as compared to the other positions, thereby enabling accurate detection of the value of the power to be detected having been input to the voice coil **103**.

[0039] In this manner, according to the speaker of the embodiment, it is possible to presume whether a failure was caused by a burnout due to an excessive input exceeding the capacity of the voice coil **103**.

[0040] Further, according to the speaker of the embodiment, it is possible to grasp the use condition of the speaker on the market by recording causes of failures.

[0041] Furthermore, according to the speaker of the embodiment, it is possible to presume whether a failure was caused by a burnout due to an excessive input, without making any changes to general production methods or structures of the speaker.

[0042] Further, according to the speaker of the embodiment, it is possible to presume whether a failure was caused by a burnout due to an excessive input, with addition of an inexpensive component.

[0043] Note that the present disclosure is not limited to the aforementioned embodiment, and can be appropriately changed within the scope without departing from the gist. For example, it is preferable that the detector **104** be disposed such that the center axis of the voice coil **103** is located on the inner side of the closed circuit of the detector **104**, but the center axis of the voice coil **103** and the center axis of the detector **104** may not be coaxial with each other. The center axis of the voice coil **103** and the center axis of the detector **104** may be displaced from each other. Further,

the plane in which the voice coil **103** is wound and the plane formed by the winding of the closed circuit of the detector **104** may be inclined.

[0044] Further, the detector **104** and the speaker **100** may be manufactured as separate bodies and the detector **104** may be fixed to the speaker **100** when the speaker **100** is used.

[0045] Furthermore, in the detector **104**, the closed circuit may be formed by, for example, forming an appearance by deposition and a single-turn coil on a specific surface of the cap **110** of the speaker **100**. A plurality of pieces of foil having different widths may be deposited on the cap of the speaker **100** so as to read the input value when the speaker **100** is damaged.

[0046] Further, the detector **104** may be attached to a sticker for a model name on a back side of the speaker **100**.

[0047] While the invention has been described in terms of several embodiments, those skilled in the art will recognize that the invention can be practiced with various modifications within the spirit and scope of the appended claims and the invention is not limited to the examples described above.

[0048] Further, the scope of the claims is not limited by the embodiments described above.

[0049] Furthermore, it is noted that, Applicant's intent is to encompass equivalents of all claim elements, even if amended later during prosecution.

Claims

1. A speaker comprising: a magnet; a vibrator; a voice coil configured to vibrate the vibrator; and at least one detector insulated from the voice coil and including a closed circuit of a conductor that is fused under a predetermined fusion condition, wherein the at least one detector is disposed at a position where a fluctuating magnetic field generated due to a current flowing to the voice coil passes through a plane formed by the closed circuit of the at least one detector, so that power is generated in the at least one detector.
 2. The speaker according to claim 1, wherein the at least one detector is disposed such that a center axis of the voice coil is located on an inner side of the closed circuit of the at least one detector.
 3. The speaker according to claim 1, wherein the at least one detector is disposed such that an angle of a plane in which the voice coil is wound and an angle of a plane formed by the closed circuit of the at least one detector are proximate to each other.
 4. The speaker according to claim 1, wherein the fusion condition is a value of power generated in the at least one detector corresponding to a maximum power value that is allowed to be input to the voice coil.
 5. The speaker according to claim 1, comprising the at least one detector including a plurality of detectors, the plurality of detectors having different fusion conditions.
 6. The speaker according to claim 1, wherein the at least one detector is fixed on an exterior surface of the speaker.
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